

Enhanced Emotion-aware Music Recommendation via Large Language Models

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Abstract

The profound psychological interplay between music and human affect underscores the critical role of emotion-aware modeling in music recommendation systems. While existing approaches predominantly employ emotion classification or vector representation methods, their capacity to capture nuanced emotional semantics remains limited. This paper presents the Large-Language-Model-based emotion-aware music recommendation model (LEMON), a novel framework that harnesses Large Language Models to address two fundamental challenges: extracting multi-faceted emotional semantics from music content and dynamically aligning users' transient emotional states with long-term preferences. Our framework introduces three key innovations: First, a hierarchical LLM-driven emotion extraction pipeline that decodes complex emotional narratives from lyrics through multi-stage prompt engineering, generating semantically rich song embeddings. Second, a dual-temporal encoder that separately models users' short-term emotion representation derived from recent engagement patterns and long-term preferences extracted from historical interactions. Third, an adaptive fusion coefficient that dynamically balances temporal features using user profile attributes to enable personalized recommendation adaptation. Extensive evaluations on industrial and public datasets demonstrate LEMON's superiority, achieving 4.22% AUC improvements over state-of-the-art baselines. Online A/B tests on NetEase Cloud Music, one of China's largest music streaming platforms

validate a 1% increase in user engagement metrics. Ablation studies confirm the necessity of each component, particularly highlighting LLMs' performance gain in capturing context-dependent emotions compared to conventional methods. This work advances the integration of LLMs' semantic reasoning capabilities into emotion-aware recommendation systems, offering practical insights for online music personalization.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Music Recommendation, Large Language Models, Emotion-aware

ACM Reference Format:

Shijia Wang, Tianpei Ouyang, Yunfan Zhou, Qiang Xiao, Yintao Ren, Yifei Pan, Fangjian Li, and Chuanjiang Luo. 2025. Enhanced Emotion-aware Music Recommendation via Large Language Models. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '25)*, August 3–7, 2025, Toronto, ON, Canada. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3711896.3737212>

1 Introduction

In recent years, with the development of music streaming platforms, the number of active users has increased dramatically, sparking significant research interest in music recommendation systems within both industrial and academic communities. Unlike other well-explored recommendation domains, music recommendation systems face unique challenges related to the nature of music. Studies have established that music is an effective form of emotion induction, capable of influencing and modulating users' emotional states[39][7]. Most existing music recommendation systems adopt methodologies from e-commerce recommendation systems, although effective, they overlook a crucial contextual factor: users' tendency to listen to music that resonates with their current emotional state[25]. This emotional congruence in music selection

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KDD '25, Toronto, ON, Canada.

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ACM ISBN 979-8-4007-1454-2/25/08
<https://doi.org/10.1145/3711896.3737212>

manifests in patterns where individuals typically gravitate toward cheerful, uplifting music during periods of happiness, and demonstrate preference for melancholic compositions during moments of sadness [16]. The emotional states of the listener in music consumption exhibit temporal patterns throughout the day [29]. In order to verify this hypothesis, we analyzed the play distribution of songs with daily play counts exceeding a certain threshold on the NetEase Cloud Music platform over five different time segments throughout the day. An interesting phenomenon was observed: songs that are widely recognized by users or explicitly labeled with certain emotional attributes in comments exhibit distinct patterns in their play distribution. In Figure 1, we present several songs with clear emotional inclinations and significant differences in their distribution patterns. For instance, songs that convey a sense of missing someone, like "IF YOU", show higher play counts during the late night and midnight hours, aligning with moments of solitude and emotional contemplation. And songs that evoke a sense of happiness, such as "APT", tend to peak in plays during the evening hours, reflecting the increased preference for cheerful music as individuals relax and enjoy their leisure time after work.

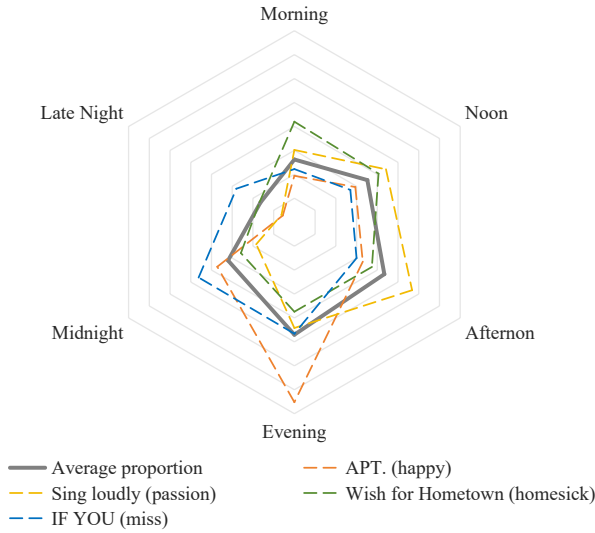


Figure 1: Time distribution of the number of songs actively played in a day. “Average” represents the time distribution of all songs actively played on the platform, while “APT” is a typical optimistic song, “Sing Loudly” is a passionate song, “Wish for Hometown” expresses the feeling of homesickness and “IF YOU” is a song about missing someone.

Existing emotion-aware music recommendation systems employ two approaches for emotional modeling of a music track. The first approach relies on the classification of emotions, utilizing a bag-of-words model to analyze the frequency of emotional words in the lyrics[5], or employing Convolutional Neural Network(CNN[17]) to classify the lyrics into specific emotional categories. The second approach involves using techniques like Doc2Vec[19] to derive a matrix for capturing and representing the emotional characteristics of a song. However, These methods do not account for the

continuity and multifaceted nature of emotional expressions in songs.

In this context, the advent of Large Language Models (LLMs) indicates a novel path for the development of recommendation systems, attributed to their rich open-world domain knowledge [21, 42]. Additionally, it brings opportunities for the advancement of emotion-driven music recommendation systems, which effectively bridge the gap between textual representations and human emotional domains [3, 23, 41]. It has demonstrated superior capabilities in textual sentiment analysis[40], offering a more sophisticated understanding of contextual nuances and subtle emotional variations compared to traditional sentiment classification models. Some music tracks can arouse people’s “nostalgia” because of their era background, which is incomprehensible to traditional recommendation systems[35, 36, 38]. Take the song *My Old Deskmate* for example:

*The sky was always blue back then
The days always passed too slowly
You always said graduation was a long way off
In the blink of an eye, we went our separate ways*

The lyrics depict a nostalgic scenario from memories, where contextual elements evoke emotional resonance among listeners, suggesting underlying sentiments of nostalgia and melancholy. LLMs are more capable of understanding such emotions. However, the application of LLMs in music emotion modeling remains largely unexplored.

While direct correlations exist between users’ emotional states and their musical preferences, these emotional influences are predominantly transient in nature[9, 28]. Consequently, to enhance user satisfaction with music recommendation, it becomes imperative to consider both short-term emotional states and long-term historical preferences[2]. Users’ musical preferences are typically represented through sequential modeling of their long-term interaction history with songs. For instance, if a user demonstrates a preference for pop music, the recommendation system typically suggests songs predominantly from this category based on interaction records. However, when the user experiences a more pleasant emotional states, his music consumption tends to shift toward more positive emotional content. However, the effective integration of short-term emotional states into this established preference framework remains a challenging research question. The state of users listening to music in a calm mood is more dependent on their long-term preferences, while emotions often affect users’ music needs when they show clear emotional fluctuations. Although existing studies have employed attention mechanisms[32] and LSTM[22], these approaches have not fully addressed the inherent heterogeneity between different feature spaces. And the matching of musical preferences and emotions is not adequately considered.

To address the above challenges, we propose a Large-Language-Model-based emotion-aware music recommendation model(LEMON). First, we employ well-designed prompts to activate LLMs to extract the emotional content contained in the lyrics. We further integrate them with music domain textual content for comprehensive song representation. Second, we introduce a dual-temporal encoder that captures users’ instantaneous emotional states and music preference by analyzing their interaction sequences. Finally, to address the alignment between users’ long-term preferences and short-term

emotions, we propose a dynamic fusion coefficient that dynamically personalizes the weights of preference and emotion representation to optimize recommendation results.

In summary, the contributions of our study are as follows:

- We propose a general emotion-aware music recommendation framework with LLMs aided. To the best of our knowledge, it is the first time that LLMs have been integrated into music emotion-aware modeling in the music recommendation scenario.
- In conjunction with a short-term emotion encoder designed for user emotion modeling, this method is integrated in a personalized manner with users' long-term preferences, thereby enhancing the accuracy of recommendations.
- Experiments conducted on multiple datasets show that the proposed method achieves state-of-the-art results.

2 Related Work

2.1 Emotion-aware Music Recommendation

Existing methods for modeling emotional attributes are often based on its inherent content information, such as textual content, audio, and other features. Naseri et al. [26] employed text-based BART [20] and audio mapping models to analyze lyrics and acoustic features, learning correlations between song emotions and lyrical-acoustic characteristics through emotion classification prediction tasks. Their findings indicate varying importance of lyrics and audio features in different emotional categories. Meanwhile, Jadhav et al. [13] suggested using BERT [6] to understand complex song emotions through lyrics and visual content representations for emotion category mapping and recommendation.

Regarding user emotional state modeling, environmental and user behavior context perception is the main approach. Deng et al. [5] leveraged users' data on social platforms to extract emotions at various granularity levels. They correlated the emotions of users, music tracks, and listening time in order to enhance the performance of music recommendations. PIEA [32] maps sentiment words from user social media posts and behavioral data to model their emotional states. Then, it integrates these modeled emotional states with users' musical preferences for song recommendations. EPMR [22] performs a low-dimensional classification mapping of the emotional characteristic of songs, constructs emotional preferences based on users' historical playback sequences, and extracts short-term emotional changes through the single music track that the user has recently liked.

2.2 LLMs with Recommendation

Utilizing LLMs as feature extractors to inject semantic features into existing recommendation systems and improve recommendation performance has emerged as a prominent research focus in recent years [24, 30, 40]. LEARN [14] constructs prompt descriptions from extracted key features as input to LLMs, generating item content embeddings enriched with open-world knowledge through the average pool of token embeddings from the final layer. Kumar et al. [18] leveraged LLMs' semantic comprehension capabilities for emotional recognition and classification by processing textual representations extracted from audio and visual features. ECR [41]

incorporates the concept of empathy to enhance conversational recommendation systems. TTMR++ [8] leverages LLMs to enhance the understanding of various song tags, thereby improving the recall performance. These studies effectively demonstrate the superior performance of LLMs in semantic feature extraction and emotional understanding. However, existing research appears to lack a comprehensive methodology for leveraging LLMs to perceive the emotional content of items and utilize this understanding to enhance the performance of recommendation systems.

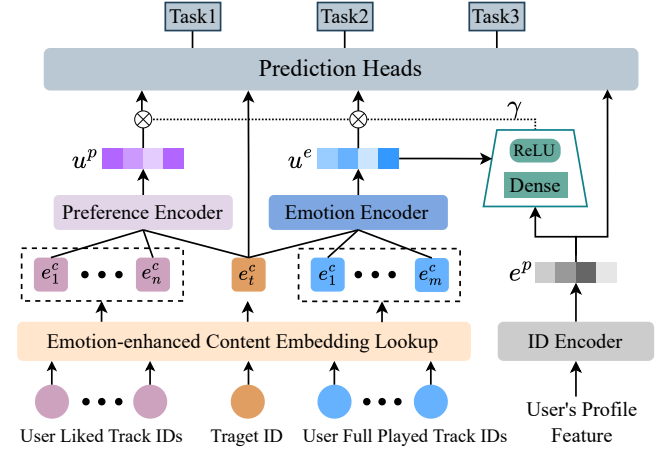


Figure 2: LLM-based emotional music recommendation framework. The emotion-enhanced content embeddings are generated via LLM-driven Emotion Extraction Pipeline and are stored offline. The γ stands for Dynamic Fusion Coefficient, which dynamically adjusts the combination of user preferences representation u^p and emotion representation u^e to optimize recommendation outcomes.

3 Methodology

3.1 The LEMON Framework

As illustrated in Figure 2, the proposed LEMON model comprises three pivotal components. Initially, we employ a hierarchical LLM-driven emotion extraction pipeline, which leverages the semantic reasoning capabilities of LLMs to decode multifaceted emotional narratives from music content, but also obviates the necessity of real-time LLM inference during the online recommendation procedure. Subsequently, in user representation modeling, we utilize a dual-temporal encoder to separately model different types of user historical interaction sequences, thereby obtaining the long-term preference representation and short-term emotion representation. Finally, we propose an adaptive fusion module that integrates user profile attributes and employs a dynamic fusion coefficient to balance the weights of long-term preferences and transient emotional states, enabling personalized and emotion-aware music recommendation.

3.2 LLM-driven Emotion Extraction Pipeline

In industrial settings, existing emotional tags for music tracks suffer from limited coverage and low accuracy. For the vast majority of

music tracks, lyrics serve as the most straightforward means to convey the emotional essence of the music track. In light of this, we introduce a context representation modeling approach centered on multi-stage interactions with LLMs. This method extracts **emotion-enhanced content embedding** with emotional attributes from the LLM via pre-inference for use by downstream recommendation models. The specific modeling process is shown in Figure 3.

In the first stage, we designed a unified prompt to drive the LLM to understand the emotional content of the lyrics. For each set of input lyrics for a song, we concatenate it with the designed prompt and then feed the combined text into the LLM to obtain the emotional summary output of the LLM. By iterating through every song in this way, we obtain the emotional summary of the lyrics for each song.

In the second stage, for each individual song, we combine the summary text of the lyric emotions obtained in the first stage with the song's metadata text information, including the song's title, artist, language, genre, and instruments. These are formed into a prompt and then input into the LLM. Then we extract the last hidden layer of the LLM, denoted $H \in \mathbb{R}^{L \times d}$, where L represents the sequence length and d is the hidden dimension. To condense the embedding representation, we apply average pooling along the sequence dimension, resulting in the emotion-enhanced content embedding $e^c \in \mathbb{R}^d$. This embedding integrates both emotional and metadata semantics, which captures the semantic and emotional essence of the lyrics. The effectiveness of e^c is rigorously validated through ablation experiments and case studies, demonstrating its capability to encode nuanced emotional narratives for downstream recommendation tasks.

3.3 Dual-temporal Encoder

After acquiring the emotion-enhanced content embeddings of all songs, we propose a **dual-temporal encoder (DTE)** to achieve a mild decoupling of the user's long-term interests and short-term emotions, thus enabling a more targeted capture of the user's emotion representation.

3.3.1 Emotion Encoder. To capture transient emotional states driven by immediate contextual factors, we design a temporally-sensitive encoder focused on recent behavioral patterns. When users exhibit emotional engagement, their listening sessions demonstrate higher levels of immersion, reflected through complete playback of emotionally resonant tracks. We construct the input sequence $S_f = \{e_1^c, e_2^c, e_3^c, \dots, e_m^c\}$ from tracks fully played within a sliding time window (e.g., 24 hours), where m denotes the temporal sequence length and each e_i^c represents the emotion-enhanced embedding of the i -th track.

The encoder employs a context-aware attention mechanism to model dynamic emotional alignment. For the target track e_t^c , we compute interaction-specific attention weights:

$$\lambda_i = \text{ReLU} \left(w_a^T \text{concat} (e_i^c, e_t^c) + b_a \right) \quad (1)$$

where w_a, b_a are learnable parameters. This design explicitly models how recent emotional interactions transiently influence current preferences. The aggregated emotional representation $u^e \in \mathbb{R}^d$ is

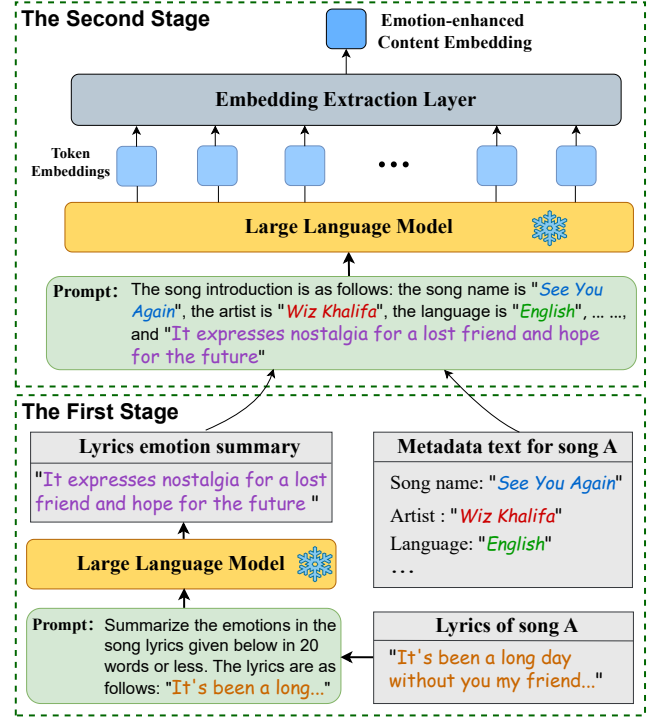


Figure 3: LLM-driven Emotion Extraction Pipeline. Stage 1: LLM generates emotion summarization from the input song lyrics. Stage 2: Combine the summary text of the lyric emotions with the metadata text of the song and extract embedding.

then derived through SoftPool [33] :

$$u^e = \sum_{i=1}^m \frac{\exp(\lambda_i) e_i^c}{\sum_j \exp(\lambda_j)} \quad (2)$$

This formulation emphasizes temporal locality and emotional volatility, distinguishing it from static preference modeling.

3.3.2 Preference Encoder. In contrast, long-term musical preferences reflect stable user interests accumulated over extended periods. We model this through explicit positive feedback signals—tracks explicitly "liked" and added to persistent collections, which is a direct and explicit manifestation of users' interests. This architecture intentionally avoids temporal constraints, instead emphasizing cross-session interest coherence and metadata-driven preference persistence. The preference sequence $S_l = \{e_1^c, e_2^c, e_3^c, \dots, e_n^c\}$ encapsulates curated musical tastes, where n represents the depth of historical interaction. The e_i^c is also the emotion-enhanced content embedding, which incorporates music metadata text, providing a comprehensive representation of the user's musical preferences.

Following a similar approach, we compute attention scores λ_i for each embedding e_i^c in the sequence S_l with respect to the embedding of the target track e_t^c , and aggregated using SoftPool:

$$u^p = \sum_{i=1}^n \frac{\exp(\lambda_i) \cdot e_i^c}{\sum_{j=1}^n \exp(\lambda_j)} \quad (3)$$

where $u^p \in \mathbb{R}^d$ represents the preference representation. This representation reflects the user’s stable musical tastes, which are less influenced by transient emotional states.

3.4 Personalized Emotion Preference Fusion

Considering the inherent disparities between preference representations and emotion representations originating from diverse distributions, simply concatenating them to form the user representation is suboptimal. Hence, we introduce a dynamic fusion coefficient to assess the degree of current emotional intensity of users. This factor takes into account the user’s profile characteristics and facilitates the dynamic modulation of the weights allocated to the preference and emotion representations within the model. Thereby, it can more accurately align with the users’ interests or emotions to conduct music recommendation.

3.4.1 Dynamic Fusion Coefficient: The Dynamic Fusion Coefficient (DFC) aims to assess the magnitude of users’ current emotional intensity, which is crucial to understanding their interaction with the Emotion-Aware Music Recommendation system. Firstly, we utilize emotion representation as a pivotal metric to quantify the extent of emotional intensity. Secondly, it is recognized that the emotional intensity of users can exhibit substantial disparities between different cohorts of users. For example, users belonging to different age brackets might display varying degrees of emotional variability. Furthermore, even within the same user group, the emotional intensity can undergo fluctuations over the course of a day. Thus, for each sample, we bin the user’s age and interaction time into non-overlapping intervals. Then, together with the user’s gender, we perform one-hot encoding on these features and concatenate the results to form the profile embedding $e^p \in \mathbb{R}^k$.

Then, we employ a gated offset function [12] to calculate the dynamic fusion coefficient γ :

$$\gamma = \sigma(w_e^T u^e + w_p^T e^p) \quad (4)$$

where $w_e \in \mathbb{R}^d$ and $w_p \in \mathbb{R}^k$ are learnable model weight parameters. The sigmoid function $\sigma(\cdot)$ is applied to the linear combination of these weighted parameters to produce a value between 0 and 1.

3.4.2 Prediction heads. Based on the dynamic fusion coefficient, we dynamically adjust the weights of u^e and u^p . Subsequently, we input the concatenated embeddings into the prediction head, which consists of the weighted u^e and u^p , the profile embedding e^p , and the emotion-enhanced content embedding of the target track e_t^c . Specifically, we employ a four-layers Multi-layer Perceptron (MLP) to integrate these embeddings, produced the logit $\hat{y}_j$ for the j -th prediction task:

$$\hat{y}_j = \text{MLP}(\gamma u^e, (1 - \gamma)u^p, e_t^c, e^p) \quad (5)$$

Although a multi-task expert network can be used to achieve better results for each task, we only use three different linear maps in the last layer of the four-layer MLP to obtain logits for multiple tasks in order to show the validity of our model.

We train the model in a multi-task learning fashion. The loss function is defined as the sum of the binary cross entropy losses of

each task, averaged by the number of training samples:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^T y_{ij} \log \sigma(\hat{y}_{ij}) + (1 - y_{ij}) \log(1 - \sigma(\hat{y}_{ij})) \quad (6)$$

where N is the total number of training instances, T is the number of the tasks, and $y_{ij} \in \{0, 1\}$ is the label for each sample in different tasks.

4 Experiments

In this section, we conduct a variety of offline experiments and online tests to evaluate the proposed method. Specifically, our aim is to address the following research questions:

- **RQ1** How does LEMON perform compared to both general music recommendation models and emotion-aware music recommendation models?
- **RQ2** How does each component of LEMON improve different recommendation tasks?
- **RQ3** Can emotion-enhanced content embedding accurately represent the emotional attributes of music tracks?
- **RQ4** How does LEMON perform in real-world industrial applications?

4.1 Experimental Settings

4.1.1 Datasets. We conducted extensive experiments on both an industrial dataset and the public Music4all[31] dataset. The core schema of all datasets includes the user’s anonymized ID and profile information, the target track ID along with its metadata text, emotion-enhanced embeddings, binary interaction labels (where positive labels indicate occurrence and negative labels indicate non-occurrence), and timestamps for temporal modeling. The statistics of these datasets are summarized in Table 1. The industrial dataset comprises 8.82 million user interaction records collected from NetEase Cloud Music platform over one week, involving 4.89 million users and 1.33 million songs. This dataset encompasses user interactions including clicks, likes, and playback completions, along with the corresponding timestamps. Music domain textual content includes important metadata of music, such as title, artist, language, genre, instruments and lyrics. The original Music4all dataset contains 5.1 million user interaction records. After constructing the user historical interaction sequences, we obtained 2.59 million interactions, covering 14,127 users and 99,596 songs. It also includes user playlists, timestamps, and textual contents of music (title, artist, album, genre, and lyrics). The Music4all dataset is split into training and test sets with a 9:1 ratio.

Table 1: Statistics of datasets

Dataset	#Users	#Songs	#Interactions
Industrial	4,895,568	1,334,065	8,823,770
Music4all	14,127	99,596	2,597,382

4.1.2 Baselines. We compare the proposed LEMON with the following generic recommendation methods and emotion music recommendation methods.

Table 2: Offline Experiment Results. The best performance is highlighted in bold and the second to best is highlighted by underlines. Improve indicates relative improvements over the best baseline in percentage.

Model	Industrial				Music4all	
	click_AUC	full_play_AUC	like_AUC	Logloss	replay_AUC	Logloss
DNN	0.5612	0.6134	0.7074	1.643	0.6004	1.149
DeepFM	0.5650	0.6161	0.7106	1.564	0.6101	1.158
DIN	0.5665	0.6151	<u>0.7206</u>	<u>1.236</u>	0.6082	<u>0.617</u>
PEIA	<u>0.5706</u>	<u>0.6177</u>	0.7156	1.604	0.6134	1.087
EPMR	<u>0.5679</u>	0.6176	0.7199	1.570	<u>0.6165</u>	1.164
LEMON	0.5801	0.6299	0.7307	1.225	0.6425	0.608
%Improve	+1.66%	+1.98%	+1.40%	-0.89%	+4.22%	-1.46%

- **DNN[34]**: This model employs a deep learning architecture with multiple layers to capture complex feature interactions. It is adept at learning representations from large datasets, making it a strong baseline for various recommendation tasks.
- **DeepFM[10]**: This model combines factorization machines with deep learning to capture both linear and non-linear feature interactions, offering a powerful approach for recommendation tasks.
- **DIN[43]**: This model utilizes attention mechanisms to dynamically capture users’ interests from their historical behaviors, enhancing the model’s ability to predict user preferences.
- **PEIA[32]**: A sophisticated model that integrates long-term personality traits and short-term emotional states of the user. It utilizes hierarchical attention mechanisms to discriminate the significance of different features when modeling the correlations between user-personality and user-emotion.
- **EPMR[22]**: It is tailored to address users’ emotional needs by recommending music that aligns with their current emotional state. It considers the dynamic nature of user emotions and aims to provide personalized music suggestions that can potentially improve users’ mental well-being.

4.1.3 Evaluation Metrics. To comprehensively evaluate the performance of our model in different tasks, we adopt the AUC (Area Under the Curve) [11] as our primary evaluation metric. Specifically, we evaluate the following three prediction tasks: (1) whether the user will click on the song (click_AUC), (2) whether the user will fully play the song (full_play_AUC), and (3) whether the user will mark the song as liked (like_AUC). These metrics reflect different aspects of user engagement and satisfaction with recommended music. In addition, under the same model parameter magnitude, we provide the Logloss metric, with the specific formula detailed in Section 3.4.2. Notably, due to the absence of behavior labels in the Music4all dataset, we conducted the repeatedly played songs as positive samples and songs played only once as negative samples, with the replay prediction task and evaluated the replay_AUC metric for this dataset.

4.1.4 Parameter Settings. For the extraction of emotion-enhanced content embeddings, we utilize two open-source LLMs tailored to the language of the datasets. For the industrial dataset, which contains Chinese lyrics, we employ the Baichuan2-7B[37] model,

quantized to 8-bit precision, to generate the embeddings. For the public Music4all dataset, which consists of non-Chinese lyrics, we use the LLaMA-3.2[1] model. Both LLMs are deployed on a NVIDIA A100-40G GPU to ensure efficient processing. It takes about tens of hours to perform the LLM-driven emotion extraction pipeline for 1.3 million songs. Based on these extracted embeddings, we train our music recommendation model with an epoch setting of 6, optimizing the model parameters to achieve the best performance in downstream recommendation tasks.

For online deployment, we conduct the computation offline and save the extracted emotion-enhanced content embeddings to KV storage system, which is further used to support large-scale deployment and online services.

4.2 Performance Comparison (RQ1)

As depicted in Table 2, we evaluate the performance of LEMON in the two data sets, comparing it with both traditional recommendation system models and state-of-the-art emotion-aware music recommendation models. The best performance is highlighted in bold, and the second to best is highlighted by underlines.

Our analysis of the offline experiment results reveals several key insights:

- (1) Compared to all baseline methods, our proposed LEMON outperforms in all prediction tasks. In the industrial dataset, click_AUC, full_play_AUC, and like_AUC are improved by 1.66%, 1.98%, and 1.40%, respectively, over the best-performing baseline. In the music4all dataset, it achieves a notable increase of 4.22% in replay_AUC. These results demonstrate the effectiveness of LEMON, as it better captures users’ emotional needs and aligns more accurately with their long-term preferences. Furthermore, on the same parameter scale, LEMON achieves the lowest Logloss, indicating its superior convergence and stability.
- (2) The performance of DeepFM and DNN is relatively poor. This can be attributed to the fact that these models are more suitable for recommendations based on collaborative information and ID embeddings. In our experiments, the items are represented solely by content embeddings, which limits the effectiveness of these methods. Additionally, both DeepFM and DNN do not specifically model the user’s historical sequence. Instead, they obtain the user representation through simple average pooling, which fails to capture the temporal dynamics and emotional

Table 3: Results of the Ablation Study

Methods	Industrial				Music4all	
	click_AUC	full_play_AUC	like_AUC	Logloss	replay_AUC	Logloss
w/o LEU	0.5784	0.6286	0.7280	1.224	0.6401	0.610
w/o DTE&DFC	0.5638	0.6142	0.7198	1.237	0.6136	0.615
w/o DFC	0.5800	0.6293	0.7291	1.225	0.6392	0.609
LEMON	0.5801	0.6299	0.7307	1.225	0.6425	0.608

context of user interactions. This lack of targeted modeling for user sequences and the reliance on average pooling for user representation contribute to their suboptimal performance.

- (3) In the experiments, PEIA, DIN and EPMR outperform DNN and DeepFM. PEIA leverages hierarchical attention mechanisms to integrate users’ long-term personality traits and short-term emotional states, enabling more discerning feature significance discrimination. DIN uses attention mechanisms to dynamically capture users’ interests from historical behaviors, enhancing preference prediction. EPMR is designed to meet users’ emotional needs by recommending music according to their current emotional state, although it does not perform attention modeling on the sequence.

These findings underscore the importance of leveraging emotional information in music recommendation and demonstrate the effectiveness of LEMON in providing personalized music suggestions that resonate with users’ emotional states and preferences.

4.3 Ablation Study (RQ2)

To verify the effectiveness of each component in LEMON, we ablated the key components of the model. And the results on the two datasets are shown in table 3, from which we can draw the following conclusions.

- (1) **w/o LLM-driven emotion understanding(w/o LEU):** In this ablation, we removed the first stage of the LLM-driven emotion extraction pipeline that focuses on summarizing the emotions in the lyrics and instead used a generic prompt to summarize the main idea in the lyrics. The second stage, which combines the lyric summarization with other song metadata, remained unchanged. This resulted in **general content embeddings** that lack the emotional enrichment provided by the LLM’s understanding of lyrical emotions and in slight decreases in AUC metrics. The performance metrics have decreased in multiple datasets, indicating that the emotional understanding prompt contributes positively to the model’s performance.
- (2) **w/o Dual-temporal Encoder & Dynamic Fusion Coefficient (w/o DTE&DFC):** This ablation experiment is mainly designed to verify the importance of the user preference representation and emotional representation obtained by modeling through differentiating different types of interaction sequences. In this experiment, we mix the sequences of songs that users have liked with the sequences of songs that users have fully played. Then, we perform representation modeling using the same attention mechanism. As a result, there is no Dynamic Fusion Coefficient to regulate the weights of the user preference representation

and the emotional representation. Across both the industrial and Music4all datasets, w/o LEU generally underperforms the LEMON model in terms of key AUC metrics and Logloss values. These results clearly demonstrate that separately modeling the long-term preference representation and short-term emotional representation can significantly improve the experimental performance. This indicates that the user’s short-term emotions are an independent contextual feature distinct from user interests, and they contribute to enhancing the effectiveness of personalized recommendation.

- (3) **w/o Dynamic Fusion Coefficient (w/o DFC):** To validate the significance of enabling the model to modulate the degree of attention allocated to the user’s long-term preferences and short-term emotions, we conducted experiments by removing the weight regulation mechanism of the Dynamic Fusion Coefficient (DFC). The experimental results indicate that while DFC has no significant impact on the user’s click prediction task, it does exert an influence on predicting user behaviors related to deep preferences, such as fully playing and liking a track. The analysis of user behavior data reveals that deviations in the user’s current listening behavior from their long-term preferences are more pronounced when users are deeply influenced by emotions. In these scenarios, the DFC mechanism allows the model to distinguish between the user’s current musical interests and emotional needs. This is crucial as it enhances the model’s adaptability to different user states, thereby improving the accuracy and relevance of personalized music recommendations.

4.4 Case Study (RQ3)

We present a case study that demonstrates the capability of emotion-enhanced content embeddings to effectively represent the emotional attributes of songs. We utilize FAISS (Facebook AI Similarity Search [15]) to perform k-Nearest Neighbors (kNN [4]) retrieval. In our experiment, we first index the emotion-enhanced content embeddings using FAISS, which partitions the embedding space into clusters and builds an inverted index for efficient retrieval. During the kNN search, FAISS computes the Euclidean distance between the query embedding and the indexed embeddings, returning the top-k most similar songs based on the smallest distances. To rigorously evaluate embedding quality, we conducted large-scale kNN retrieval on 0.1M music tracks. Emotion-enhanced embeddings show 63.4% emotion tag matching accuracy versus 48.1% for standard embeddings, confirming their superior semantic alignment with emotional labels. The specific case is as follows.

In Figure 4 (a), we illustrate the three top songs closest to the query song within the space of emotion-enhanced content embeddings. The retrieval outcomes align closely with the emotion conveyed by the target song, both expressing themes of loneliness and sorrow. This shows that emotion-enhanced embeddings effectively capture the nuanced emotional semantics of music. On the other hand, in Figure 4 (b), we show the top three similar songs retrieved for the query song using general content embeddings of music. It is evident that these retrieved songs have different emotion tags with the query song, and even exhibit completely opposite emotions. This highlights the limitations of general content embeddings in capturing emotional nuances and underscores the importance of emotion-enhanced representations for music recommendation tasks.

Through this case study, we validate the effectiveness of emotion-enhanced content embeddings, which appear to show impressive superiority in capturing context-dependent emotional semantics.

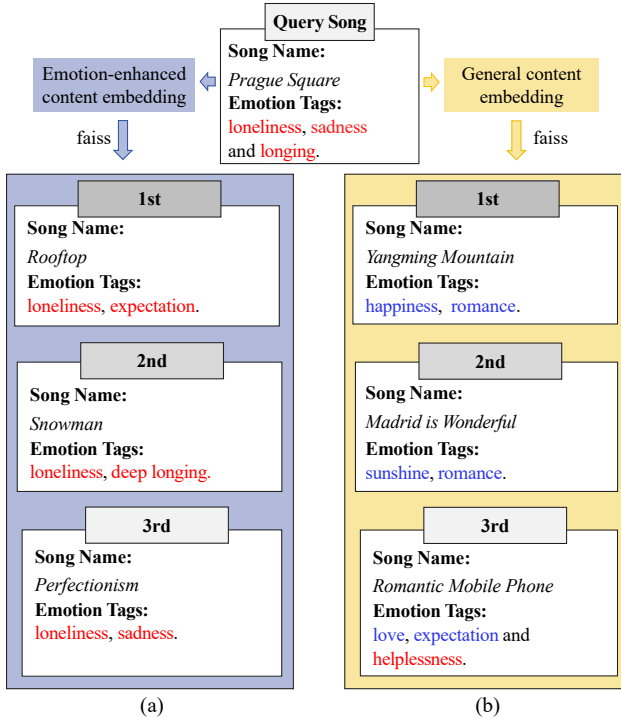


Figure 4: Retrieval results of a song. The red color marks the words with the same sentiment as the query song, and the blue color marks the words with the opposite sentiment. (a) Similarity search results using LLM emotion summary (b) Similarity search results without LLM emotion summary

4.5 Online A/B Testing (RQ4)

We deploy the proposed LEMON in the online music recommendation system of the NetEase Cloud Music platform for A/B testing and compare it with the effects of the online baseline model. The benchmark model was the industry-standard Deep Learning Recommendation Model (DLRM[27]), which integrates heterogeneous

features like user profiles, historical interactions, and collaborative filtering signals. We use like rate and play duration metrics to evaluate the performance of the model in the online A/B tests. The experimental results are presented in Table 4.

Table 4: Results of Online A/B Testing

Method	Like Rate	Play Duration
LEMON	+1%	+1.5%

Compared to benchmark models, LEMON incorporates emotion-enhanced content embeddings and introduces a dynamic fusion coefficient to modulate weights between long-term preference and short-term emotion representation. Compared to the control group, user feedback from the treatment group underscores superior mood matching, as evidenced by comments such as “perfect for late-night studying” and “no more jarring mood shifts”. Online experimental metrics further demonstrate the effectiveness of LEMON and the significance of incorporating instantaneous emotion representation to improve the performance of music recommendation.

5 Conclusion

In conclusion, we propose a novel LLM-based emotional music recommendation method LEMON, which captures emotion representations from the lyrics of songs by leveraging the deep emotional understanding capabilities of LLMs. In addition, we introduce a short-term emotion encoder to capture users’ transient mood, achieving personalized matching with long-term preferences through dynamic fusion coefficient-based weight modulation. The LEMON not only leverages the rich semantic understanding capabilities of LLMs, but also innovates the way to model user preferences and emotional needs in recommendation systems. Experiments conducted on a large-scale dataset built on the NetEase Cloud Music platform and a public dataset validated our approach and demonstrated the importance of dynamically combining users’ long-term music tastes with short-term emotional states in music recommendations. In our future research, the primary focus will be on integrating acoustic features into multimodal emotion modeling.

6 Acknowledgments

This research was supported by the Natural Science Foundation of Zhejiang Province under Grant No.LZ25F020010.

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